UNIT IV

Scalable machine learning
and
Machine learning & IoT

Outline

- **Scalable Machine Learning** :
- Machine Learning: Algorithms Types Revisited and modified, **Semi-Supervised Machine Learning**, When Can **Semi-Supervised Learning Work**?, **Active (Machine) Learning**, **Graphical Model**, **Inference on Graphical Models**, **Probabilistic Graphical Models (PGM).**
- **Machine Learning & IoT :**
- **Internet of Things, Emergence of Internet of Things, The Architecture of IoT,** Machine Learning Algorithm for IoT, **Internet of Things Communication Protocols, The IoT Architectural Reference Model, Taxonomy of Machine Learning Algorithms**, Research Trends and Open Issues.

Algorithms types revisited and modified

- The Machine Learning algorithms can be broadly classified into three categories, **Supervised Learning, Unsupervised Learning** and **Reinforcement Learning.**
- Casting Reinforcement Learning aside, the primary two categories of Machine Learning problems are Supervised and Unsupervised Learning.
- The basic difference between the two is that Supervised Learning datasets have an output label associated with each tuple while Unsupervised Learning datasets do not.
- The most basic disadvantage of any Supervised Learning algorithm is that the dataset has to be hand-labeled either by a Machine Learning Engineer or a Data Scientist.
- This is a very *costly process*, especially when dealing with large volumes of data. The most basic disadvantage of any **Unsupervised Learning** is that it's **application spectrum is limited**.

Semi-Supervised Machine Learning

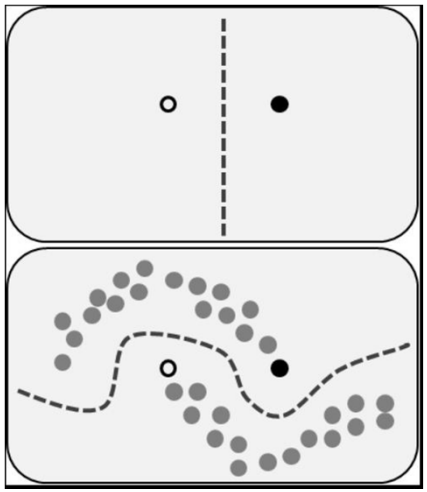
- To counter these disadvantages, the concept of **Semi-Supervised Learning** was introduced.
- In this type of learning, the algorithm is trained upon a combination of labeled and unlabeled data.
- Typically, this combination will contain a very small amount of labeled data and a very large amount of unlabeled data.
- The basic procedure involved is that first, the programmer will cluster similar data using an unsupervised learning algorithm and then use the existing labeled data to label the rest of the unlabeled data.
- The typical use cases of such type of algorithm have a common property among them – The acquisition of unlabeled data is relatively cheap while labeling the said data is very expensive.

Semi-Supervised Machine Learning

- Naturally, one may imagine the three types of learning algorithms as **Supervised learning** where a student is under the supervision of a teacher at both home and school, **Unsupervised learning** where a student has to figure out a concept himself and **Semi-Supervised learning** where a teacher teaches a few concepts in class and gives questions as homework which are based on similar concepts.

Semi-Supervised Machine Learning

- An example of the influence of unlabeled data in semi-supervised learning.
- The top panel shows a decision boundary that might adopt after seeing only one positive (white circle) and one negative (black circle) example.
- The bottom panel shows a decision boundary that might adopt if, in addition to the two labeled examples, we were given a collection of unlabeled data (gray circles).
- This could be viewed as performing clustering and then labeling the clusters with the labeled data, pushing the decision boundary away from high-density regions, or learning an underlying one-dimensional manifold where the data reside.



Semi-Supervised Machine Learning

- A Semi-Supervised algorithm assumes the following about the data –

1. **Continuity Assumption:** The algorithm assumes that the points which are closer to each other are more likely to have the same output label.
2. **Cluster Assumption:** The data can be divided into discrete clusters and points in the same cluster are more likely to share an output label.
3. **Manifold Assumption:** The data lie approximately on a manifold of much lower dimension than the input space. This assumption allows the use of distances and densities which are defined on a manifold.

Applications of Semi-Supervised Learning

1. **Speech Analysis:** Since labeling of audio files is a very intensive task, Semi-Supervised learning is a very natural approach to solve this problem.
2. **Internet Content Classification:** Labeling each webpage is an impractical and unfeasible process and thus uses Semi-Supervised learning algorithms. Even the Google search algorithm uses a variant of Semi-Supervised learning to rank the relevance of a webpage for a given query.
3. **Protein Sequence Classification:** Since DNA strands are typically very large in size, the rise of Semi-Supervised learning has been imminent in this field.

Active Learning

- **What is Active Learning?**
- Active Learning is a special case of Supervised Machine Learning.
- This approach is used to construct a high performance classifier while keeping the size of the training dataset to a minimum by actively selecting the valuable data points.
- **Where should we apply active learning?**

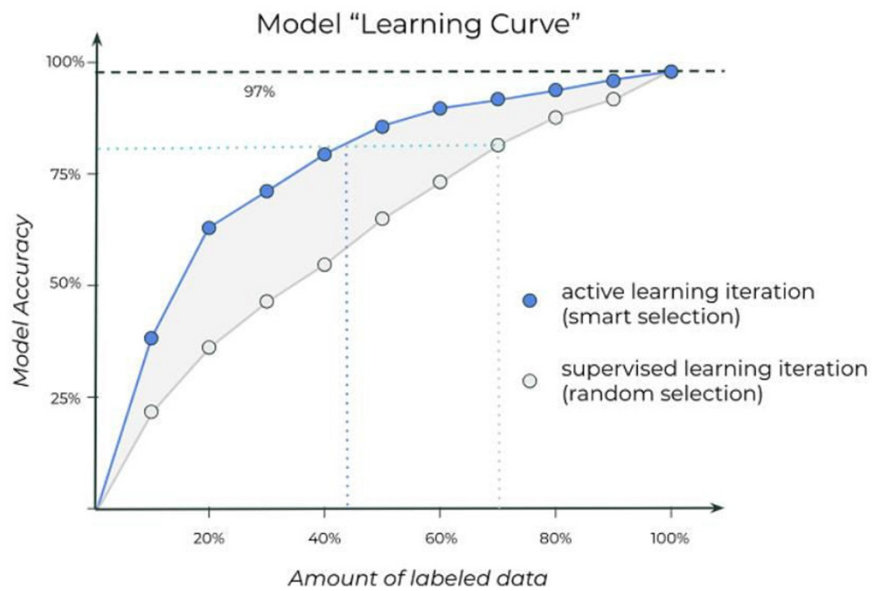
1. For a very small amount or a huge amount of dataset.
2. Annotation of the unlabeled dataset cost human effort, time and money.
3. When user have access to limited processing power.

Active Learning

- Active Learning is actually surprisingly well-motivated in many modern ML problems, in particular when labels are difficult, time-consuming, or expensive to collect.
- In Active Learning, the learning algorithm is allowed to proactively select the subset of available examples to be labeled next from a pool of yet unlabeled instances.
- The fundamental belief behind the concept is that a Machine Learning algorithm could potentially achieve a better accuracy while using fewer training labels if it were allowed to choose the data it wants to learn from. Such an algorithm is referred to as an **active learner**.
- Active learners are allowed to dynamically pose queries during the training process, usually in the form of unlabeled data instances to be labeled by what is called an **oracle**, usually a human annotator. As such, active learning is one of the most powerful examples of the success of the Human-in-the-Loop paradigm.

Active Learning

Figure 2: As expected, the more data available to train a model, the higher the accuracy regardless of whether or not an Active Learning approach is used. However, with Active Learning, a given accuracy can be reached with significantly less data; in this example, a 80% accuracy can be achieved with only 45% of the total volume of available data as opposed to 70% in the case of regular supervised learning.



How does Active Learning work?

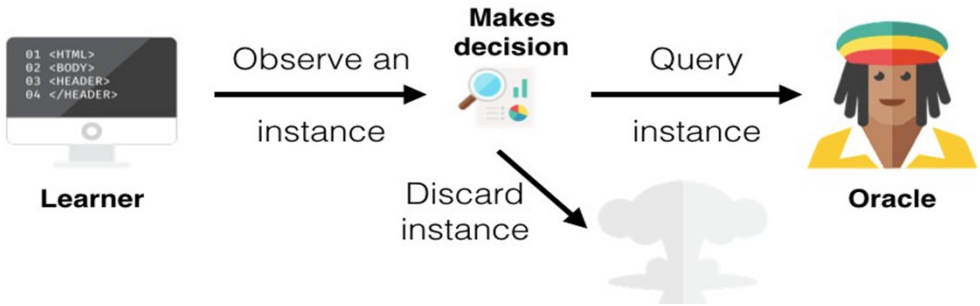
- Active Learning can be implemented through different **scenarios**.
- In essence, deciding whether or not to query a specific label comes down to deciding whether the gain from obtaining the label offsets the cost of collecting that information.
- In practice, making that decision can take several forms depending on whether the scientist has a limited budget or simply tries to minimize his labeling bill.
- Overall, three different categories of active learning can be listed:
  - **Membership query synthesis scenario**
  - **Stream-based selective sampling scenario**
  - **Pool-based sampling scenario**

Membership Query Synthesis

- This is a big term which simply means that the learner generates/constructs an instance (from some underlying natural distribution).
- For example, if the data is pictures of digits, the learner would create an image that is similar to a digit and this created image is sent to the oracle to label.

Stream-Based Selective Sampling

- In this setting, you make the assumption that getting an unlabelled instance is free.
- Based on this assumption, you then select each unlabelled instance one at a time and allow the learner to determine whether it wants to query the label of the instance or reject it based on its informativeness.
- To determine informativeness of the instance, you use a query strategy .
- Following with the example above, you would select one image from the set of unlabelled images, determine whether it needs to be labelled or discarded, and then repeat with the next.



Pool-Based sampling

- This setting assumes that there is a large **pool** of unlabelled data, as with the stream-based selective sampling.
- Instances are then drawn from the pool according to some informativeness measure.
- This measure is applied to all instances in the pool (or some subset if the pool is very large) and then the most informative instance(s) are selected.
- This is the most common scenario in the active learning community.
- Continuing with the example in the above two scenarios, all the unlabelled images of digits will be ranked and then the best (most informative) instance(s) will be selected and their labels requested.

Graphical Model

- The Graphical model (GM) is a branch of ML which uses a graph to represent a domain problem.
- Many ML & DL algorithms, including ---
  - Naive Bayes' algorithm,
  - Hidden Markov Model
  - Restricted Boltzmann machine
  - Neural Networks belong to the Graphical model .

Probabilistic Graphical Models (PGM)

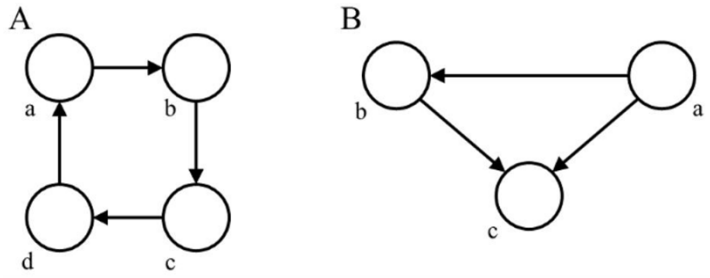
- **Probabilistic Graphical models (PGMs)** are statistical models that encode complex joint multivariate probability distributions using graphs.
- In other words, PGMs capture conditional independence relationships between interacting random variables.
- This is beneficial since a lot of knowledge on graphs has been gathered over the years in various domains, especially on separating subsets and functions on graphs.
- This knowledge can be reused in PGMs.
- Furthermore, one can easily visualize PGMs and get a quick overview of the model structure.

Probabilistic Graphical Models (PGM)

- By knowing the graph structure of a PGM, one can solve tasks such as **inference** (computing the marginal distribution of one or more random variables) or **learning** (estimating the parameters of probability functions).
- One can even try to learn the structure of the graph itself, given some data.
- Here a concise description on the two flavors of PGMs is given:
  - **Directed Graphical Models (DGMs),** known as **Bayesian Networks (BNs)**
  - **Undirected Graphical Models (UGMs) or Markov Random Fields (MRFs)**.

Directed Graphical Models (DGMs)

- As the name already suggests, directed graphical models can be represented by a graph with its vertices serving as random variables and **directed** edges serving as dependency relationships between them



- The **direction of the edges** determines the **influence of one random variable on another**.
- If the graph does not contain cycles (a number of vertices connected in a closed chain), it is usually referred to as a **Directed Acyclic Graph (DAG)**.
- Inference on these graphs may be performed exactly using algorithms such as **Belief Propagation (BP)** or variable elimination.

Bayesian Networks (BNs)

- An example of a DGM is the **Bayesian network (BN)**.
- The Bayesian Network is a **DAG** with vertices (random variables) representing observable or latent variables of the model.
- The **directed edges** ("arrows") of a BN **represent conditional distributions**.
- If the values of the vertices are binary, for example, the conditional distributions may be Bernoulli distributions.
- In case of continuous values, the conditional distributions may be Gaussian.
- The joint probability distribution is formulated as a product of conditional or marginal probabilities.

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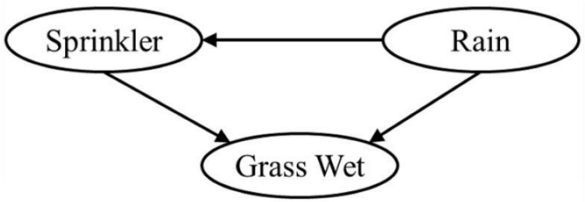
Bayesian Networks (BNs)

- For instance, when modelling the probability of wet grass given if it is raining or if the sprinkler is on, we might represent it using a DAG such as this one.



- This DAG represents the (factorized) probability distribution

$$p(S, R, G) = p(R)p(S \mid R)p(G \mid S, R)$$

- where R is the random variable for rain, S for the sprinkler and G for the wet grass. By examining the graph, you quickly see that the only independent variable in the model is R.

- The other two variables are conditioned on the probability of rain and/or the sprinkler.
- BNs are often used to model causal relationships.
- If the direction of edges in a BN represents causality, the causal allows one to modify the graph by performing a simulated intervention and predict the impact of an external interventions .

Inference In BNs

- Let us return to the wet grass example and figure out the probability that the grass is wet, i.e. **the marginal probability** *p(G)*. This task is known as inference.
- In order to determine the marginal probability of a variable, you must in general sum over all its parents in the graph.
- In this case, the marginal probability would be

$$p(G) = \sum_R \sum_S p(S, R, G) = \sum_R \sum_S p(R)p(S \mid R)p(G \mid S, R)$$

- Here we need to sum over all the factors. However, for more complicated graphs, it might be possible to **pull some factors out of the sum and simplify the calculation** greatly, making use of the **sparse graph structure**.

Inference In BNs

- Another task we may be interested in is to calculate the **conditional probability** that the grass will be wet given that it is not raining.
- In this case, we would proceed as follows

$$p(G|R) = \sum_S p(S, G \mid R) = \sum_S p(S \mid R) p(G \mid S, R)$$

- In this case we only have to marginalize over S since R is already assumed as given.
- This procedure is called Variable elimination. Variable elimination is an exact inference algorithm.
- It can also be used to figure out the state of the network that has maximum probability by simply exchanging the sums by max functions.
- Its downside is that for large BNs it might be computationally intractable.

Undirected Graphical Models (UGMs) Or Markov Random Fields (MRFs)

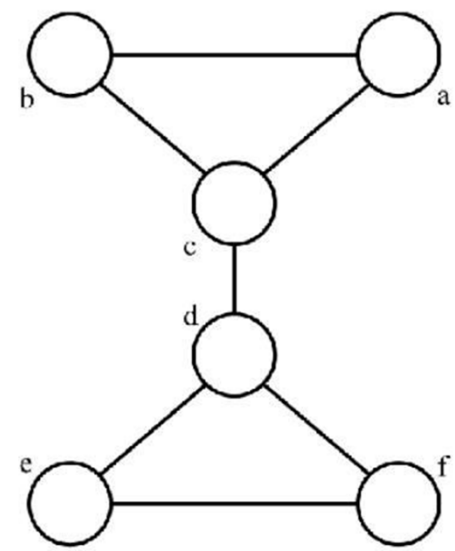
- Similar to Bayesian networks, MRFs are used to describe dependencies between random variables using a graph. However, **MRFs use undirected instead of directed edges**.
- They may also contain **cycles**, unlike Bayesian Networks.
- Thus, MRFs can describe a different set of dependency relationships than their Bayesian network counterparts.
- Note that MRFs are not a superset of DGMs since there are relationships (like causal relationships) that can only be described by DGMs.
- While the edges of Bayesian networks represent conditional dependencies, the undirected edges of **MRFs represent joint probabilities of cliques** in the graph.

Undirected Graphical Models (UGMs) Or Markov Random Fields (MRFs)

- Take a look at the following MRF:
- This graph describes a joint probability function of 6 variables in 3 distinct cliques.
- Thus it factorizes into the following product of distributions:

$$p(X) = p(X_a, X_b, X_c)p(X_c, X_d)p(X_d, X_e, X_f)$$

- A fundamental property of MRFs is that they satisfy the **pairwise, local and global Markov properties**.
- These properties share a deep connection with the d-separation rules for BNs.
- In particular, d-separation defines the Markov properties on directed graphs.



Undirected Graphical Models (UGMs) Or Markov Random Fields (MRFs)

- **The pairwise Markov property** states that two non-neighboring variables are conditionally independent given all other variables:

$$X_a \perp X_b \mid X_{G \setminus \{a,b\}}$$

, with *X_a* and *X_b* defining any two non-neighboring variables and X_*G* being the set of all variables.

- **The local Markov property** introduces the concept of the **neighborhood** of a variable:

$$X_a \perp X_{G \setminus N(a)} \mid X_{N(a)}$$

, where *N(a)* is the neighborhood of *X_a*. In other words, any variable is conditionally independent of any other variables given its neighborhood.

Undirected Graphical Models (UGMs) Or Markov Random Fields (MRFs)

- Finally, the **global Markov property** states that any set of variables *X_A* is independent of any other set *X_B* given a separating subset *X_S*:

$$X_A \perp X_B \mid X_S$$

- In general, the pairwise Markov property follows from the local Markov property, which in turn follows from the global Markov property.
- However, for strictly positive probability distributions, the statements are equivalent.

Machine Learning & IoT

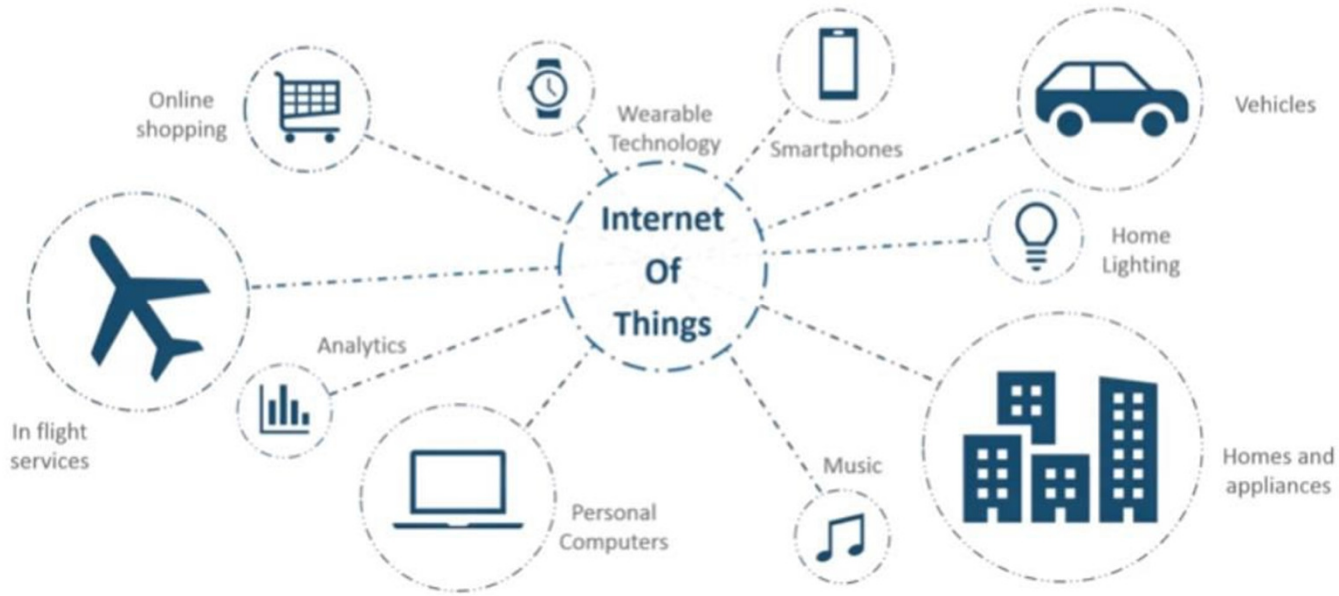
- Internet of Things
- Emergence of Internet of Things
- The Architecture of IoT
- Machine Learning Algorithm for IoT
- Internet of Things Communication Protocols
- The IoT Architectural Reference Model
- Taxonomy of Machine Learning Algorithms
- Research Trends and Open Issues.

Internet of Things

- IoT stands for Internet of Things, which means accessing and controlling daily usable equipment's and devices using Internet.
- Connecting everyday things embedded with electronics, software, and sensors to internet enabling to collect and exchange data without human interaction called as the Internet of Things (IoT).
- The term "Things" in the Internet of Things refers to anything and everything in day to day life which is accessed or connected through the internet.

- Let's us look closely at our mobile device which contains GPS Tracking, Adaptive brightness, Voice detection, Face detection etc.
- These components have their own individual features, but what about if these all communicate with each other to provide a better environment?
- For example, the phone brightness is adjusted based on my GPS location or my direction.

What is an Internet of Things (IoT)



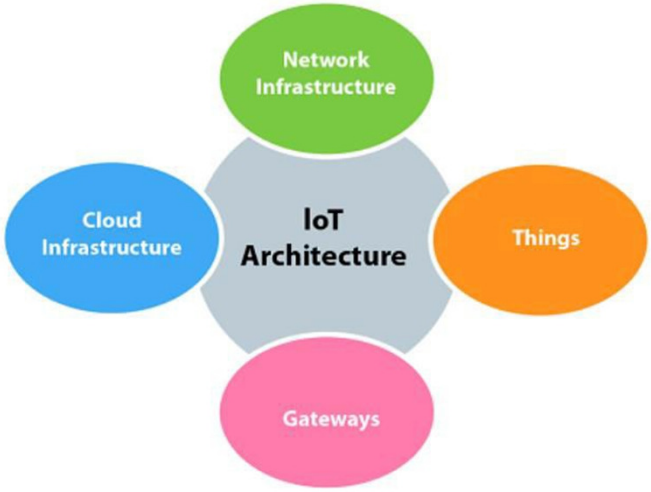
What is an Internet of Things (IoT)

- IoT is an advanced automation and analytics system which deals with artificial intelligence, sensor, networking, electronic, cloud messaging etc. to deliver complete systems for the product or services.
- The system created by IoT has greater transparency, control, and performance.
- As we have a platform such as a cloud that contains all the data through which we connect all the things around us.

- For example, a house, where we can connect our home appliances such as air conditioner, light, etc. through each other and all these things are managed at the same platform. Since we have a platform, we can connect our car, track its fuel meter, speed level, and also track the location of the car.

What is an Internet of Things (IoT)

- If there is a common platform where all these things can connect to each other would be great.
- For example, if you want room temperature to to be set at 25 or 26-degree Celsius when you reach back home from your office, then according to your car location, your AC would start before 10 minutes you arrive at home.
- This can be done through the Internet of Things (IoT).
- The working of IoT is different for different IoT echo system (architecture).
- The entire working process of IoT starts with the device themselves, such as smartphones, digital watches, electronic appliances, which securely communicate with the IoT platform.
- The platforms collect and analyze the data from all multiple devices and platforms and transfer the most valuable data with applications to devices.



Emergence of Internet of Things

- The emergence of the internet has completely changed the way we lead our lives.
- After internet, if there's any other such a consequential invention that has happened it has to be the Internet of Things.
- Though the seed of IoT was sown during the year 1982 with a modified Coke machine at the Carnegie Mellon University, it is only in last few years that it has caught people's eye.
- Still, in its initial phase, IoT is anticipated of being the future of technology. Nowadays, the Internet of things has become an integral part of many human's lives.
- Due to the emergence of wireless networks and the introduction of new connected devices people have surrounded themselves with a network infrastructure, which in turn solves the problems that previously had to be solved independently.
- According to a large number of forecasts from analysts, with the growth of connected devices the concept of the "Internet of things" will expand.

Emergence of Internet of Things

- At the moment, fifteen years after the appearance of IoT, the Internet of things has become among the most popular high-tech technologies. Probably in the very near future the boundaries between the Internet of things and the regular Internet will disappear completely and there will be something new and universal.
- Today, a virtual augmentation has come with the emergence of IoT in different fields. IoT is a technological ecosystem built to let the physical objects collect and transfer data with the help of internet network without any manual assistance.
- Basically, the smart devices interact and share data with each other, remotely and with any intervention of the human.
- IoT is all about making a smart living without much of human efforts.
- The emergence of IoT is going to come with a host of transformational changes.
- It will change the way we look and think about technology. Amongst a list of sectors that it is going to bring an impact upon, we've here picked the ones that are going to have the most prominent effect of IoT.

Emergence of Internet of Things

- **IoT & Human Lives:**
- Any form of technology has been extensively making a huge impact on human lives, especially the ones associated with the internet.
- And IoT is one of a major player that is going to frame how the human world is going to perform in future.
- Energy saving and pollution control will remain at the base of it.
- Here are some of the influences that are witness with the emergence of IoT.
- Smart homes will come into existence, where technology will take care of your daily tasks. A lot of things like electricity and appliances will be automated and synced well to make our lives easy and convenient.
- Smart cities will be constructed, where everything from schools to supermarkets, transportation, traffic update, street lighting, the collection of garbage, pollution control ,etc. will be updated with technology.

Emergence of Internet of Things

- **IoT & Business:**
- A business can't survive and grow if it's not in sync with the trending and ever-changing world.
- And as foreseen by the top technocrats of the world, IoT is going to have a deep influence on the way businesses are run today.
- Inventory Management, Human Resource Management, Projects Management, Data Management, etc. will be digitized with the emergence of IoT, which will save time and bring in more concrete organization and accuracy.
- Meetings, Presentations, Interviews, et al will not require physical presence anymore.
- A host of such activities will be done online, which will eventually save more time wasted on commuting from one place to another and that time can be allotted for more productive activities.

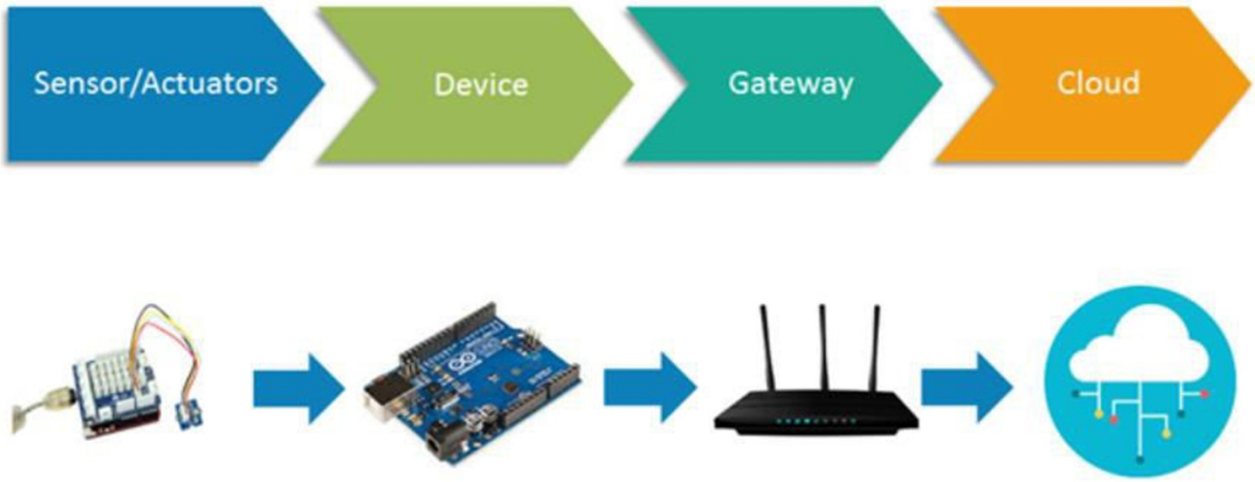
Emergence of Internet of Things

- **IoT & Business:**
- We never knew that the internet is going to become such an integral part of our living when it was invented.
- But today, it has penetrated every possible aspect of our lives, from how we interact/network with people to how we do the business, run the education system, healthcare sector, et al.
- Similarly, IoT too is going to be impacting all such factors at large. The phase of a big change has begun with IoT.

The Architecture of IoT

- There is not such a unique or standard agreement on the Internet of Things (IoT) architecture which is universally defined.
- The IoT architecture differs from their functional area and their solutions. However, the IoT architecture technology mainly consists of four major components:
- **Components of IoT Architecture**
  - Sensors/Devices
  - Gateways and Networks
  - Cloud/Management Service Layer
  - Application Layer

The Architecture of IoT



Stages of IoT Architecture

- There are several layers of IoT built upon the capability and performance of IoT elements that provides the optimal solution to the business enterprises and end-users.
- The IoT architecture is a fundamental way to design the various elements of IoT, so that it can deliver services over the networks and serve the needs for the future.
- Following are the primary stages (layers) of IoT that provides the solution for IoT architecture.
- **1. Sensors/Actuators:** Sensors or Actuators are the devices that are able to emit, accept and process data over the network. These sensors or actuators may be connected either through wired or wireless. This contains GPS, Electrochemical, RFID, etc. Most of the sensors need connectivity through sensors gateways. The connection of sensors or actuators can be through a Local Area Network (LAN) or Personal Area Network.

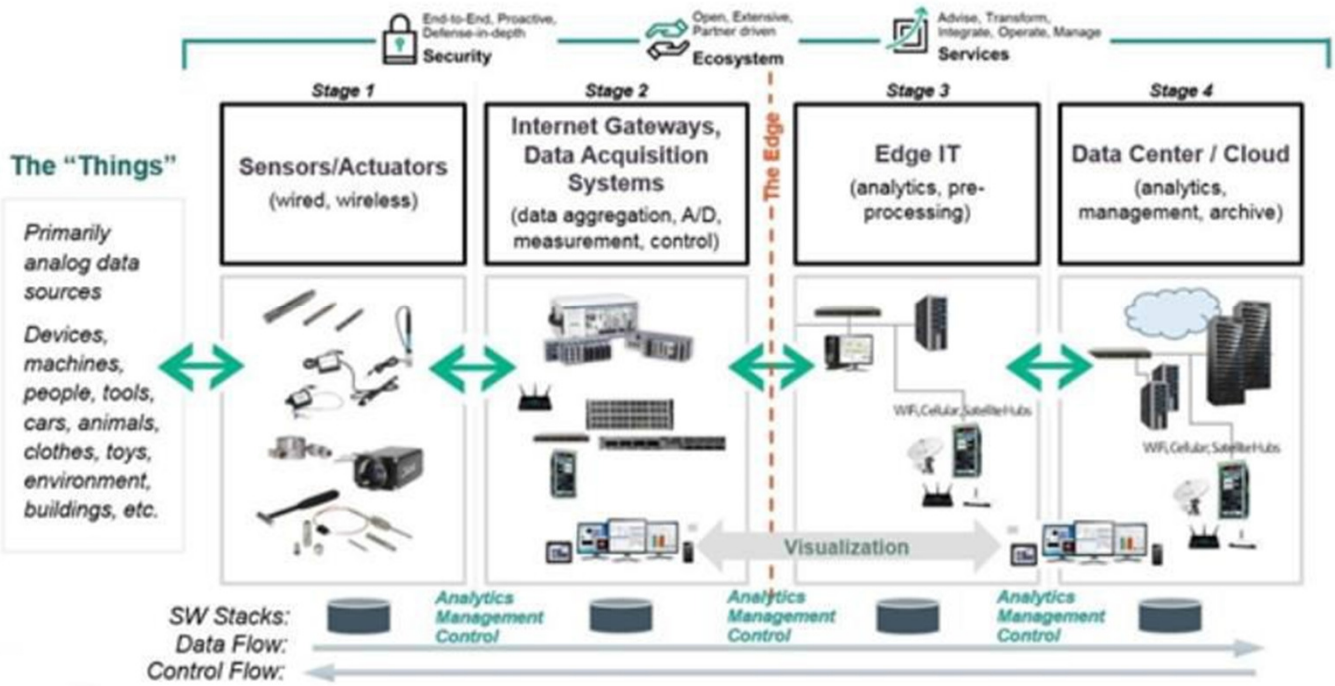
Stages of IoT Architecture

2.Gateways and Data Acquisition: As the large numbers of data are produced by this sensors and actuators need the high-speed Gateways and Networks to transfer the data. This network can be of type Local Area Network (LAN such as WiFi, Ethernet, etc.), Wide Area Network (WAN such as GSM, 5G, etc.).

3.Edge IT: Edge in the IoT Architecture is the hardware and software gateways that analyze and pre-process the data before transferring it to the cloud. If the data read from the sensors and gateways are not changed from its previous reading value then it does not transfer over the cloud, this saves the data used.

4.Data center/ Cloud: The Data Center or Cloud comes under the Management Services which process the information through analytics, management of device and security controls. Beside this security controls and device management the cloud transfer the data to the end users application such as Retail, Healthcare, Emergency, Environment, and Energy, etc.

The 4 Stage IoT Solutions Architecture



Internet of Things Communication Protocols

- The Internet of things is based on the networking of things.
- In a nutshell, Internet of Things is defined as a " *proposed development of the internet in which everyday objects have network connectivity, allowing them to send and receive data '*.
- Several Communication Protocols are used in the Internet of Things.
- Some of the major IoT protocol (IoT Communication Protocols) are Bluetooth, Zigbee, Wi-Fi, Cellular, NFC and Wi-Fi Direct.
- These IoT communication protocols provide and meet the specific functional requirement of an IoT system.

Internet of Things Communication Protocols

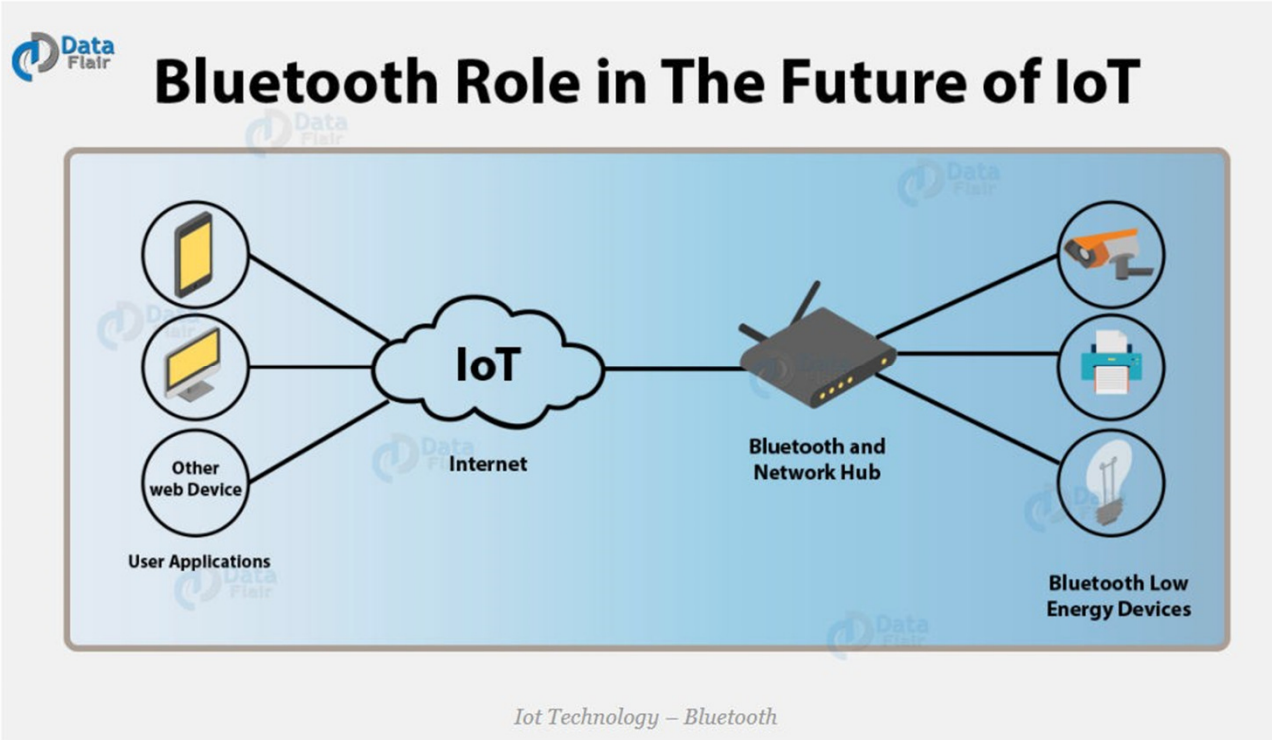
- There are various IoT Communication Protocols/ Technology

1. Bluetooth
2. Zigbee
3. Wi-Fi
4. Cellular
5. NFC (Near Field Communication)
6. LoRaWAN
7. Z-Wave

Bluetooth

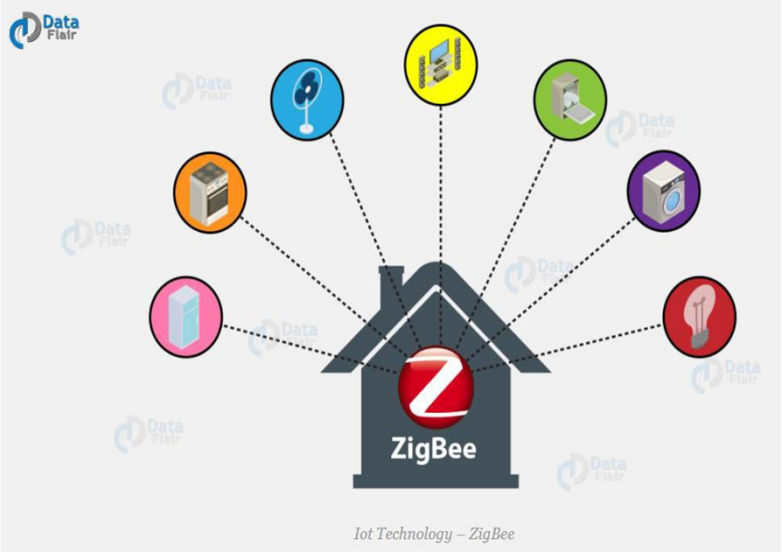
- It is an important short-range IoT communications Protocols.
- Bluetooth, which has become very important in computing and many consumer product markets.
- It is expected to be key for wearable products in particular, again connecting to the IoT probably via a smartphone in many cases.
- The new Bluetooth Low-Energy (BLE) – or Bluetooth Smart, as it is now branded – is a significant protocol for IoT applications.
- Importantly, while it offers a similar range to Bluetooth it has been designed to offer significantly reduced power consumption.

Bluetooth


Data Flair
Bluetooth Role in The Future of IoT
IoT
Internet
Other web Device
User Applications
Bluetooth and Network Hub
Bluetooth Low Energy Devices
Iot Technology – Bluetooth

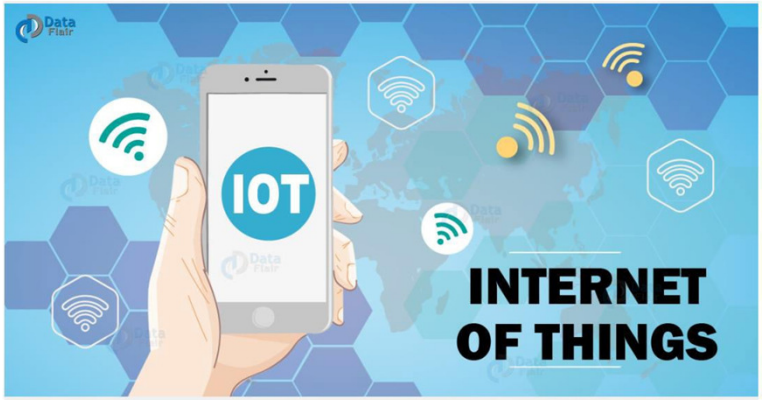
Zigbee

- ZigBee is similar to Bluetooth and is majorly used in industrial settings.
- It has some significant advantages in complex systems offering low-power operation, high security and robustness.
- It is well positioned to take advantage of wireless control and sensor networks in IoT applications.
- The latest version of ZigBee is the recently launched 3.0, which is essentially the unification of the various ZigBee wireless standards into a single standard.



WiFi

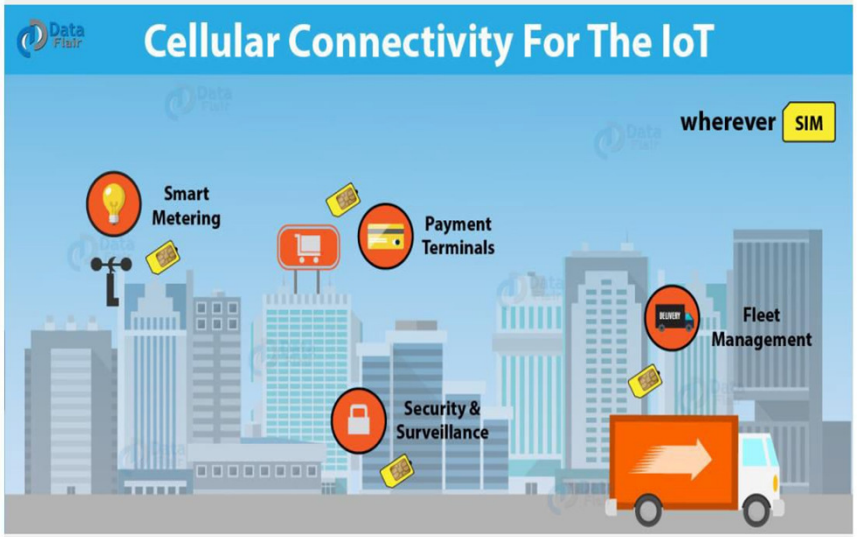
- WiFi connectivity is one of the most popular IoT communication protocol, often an obvious choice for many developers, especially given the availability of WiFi within the home environment within LANs.
- There is a wide existing infrastructure as well as offering fast data transfer and the ability to handle high quantities of data.
- Currently, the most common WiFi standard used in homes and many businesses is 802.11, which offers range of hundreds of megabit per second, which is fine for file transfers but may be too power-consuming for many IoT applications.



IoT Communication Protocols – Wi-Fi

Cellular

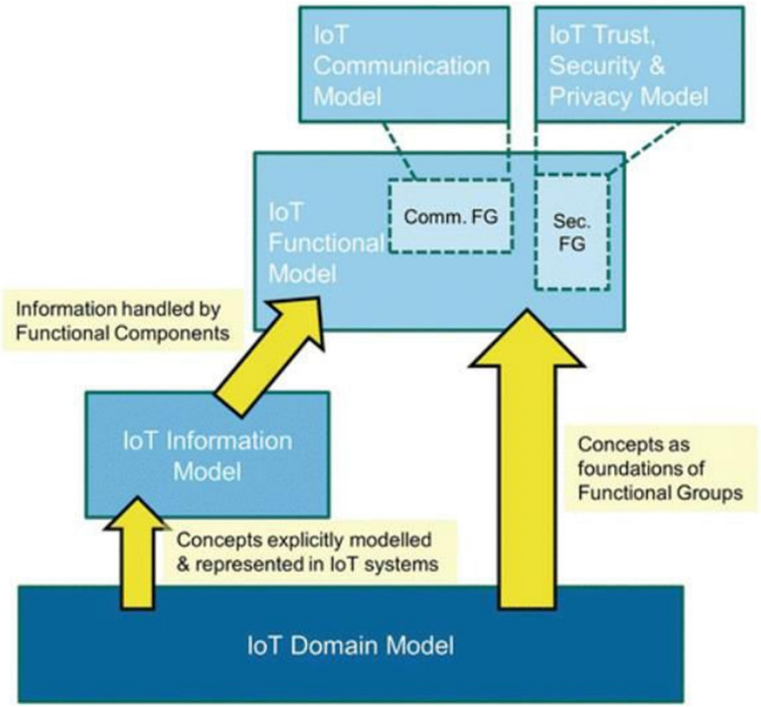
- Any IoT application that requires operation over longer distances can take advantage of GSM/3G/4G cellular communication capabilities.
- While cellular is clearly capable of sending high quantities of data, especially for 4G, the cost and also power consumption will be too high for many applications.
- But it can be ideal for sensor-based low-bandwidth-data projects that will send very low amounts of data over the Internet.



IoT Communication Protocols – Cellular

The IoT Architectural Reference Model (ARM)

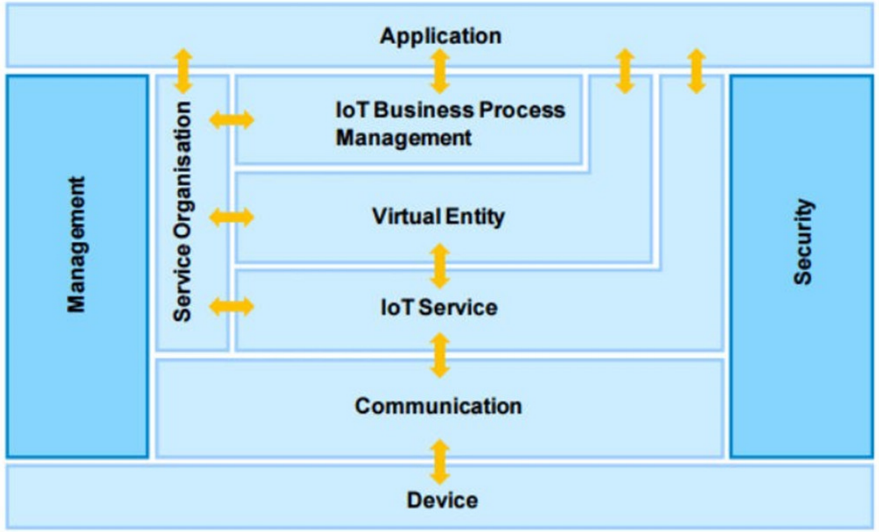
- An ARM consists of two main parts: a Reference model and a Reference Architecture.
- A reference model describes the domain using a number of sub-models



- ***IoT domain model***
  - The domain model captures the basic attributes of the main concepts and the relationship between these concepts.
  - A domain model also serves as a tool for human communication between people working in the domain in question and between people who work across different domains.
- ***Information Model***
  - Virtual Entity in the IoT Domain Model is the "Thing" in the Internet of Things, the IoT information model captures the details of a Virtual Entity- centric model.
  - Similar to the IoT Domain Model, the IoT Information Model is presented using Unified Modelling Language (UML) diagrams.

• *Functional model*

- The IoT Functional Model aims at describing mainly the Functional Groups (FG) and their interaction with the ARM.
- While the Functional View of a Reference Architecture describes the functional components of an FG, interfaces, and interactions between the components.
- The Functional View is typically derived from the Functional Model in conjunction with high-level requirements.



- ***Communication model***

- ***Safety***
- the IoT Reference Model can only provide IoT-related guidelines for ensuring a safe system to the extent possible and controllable by a system designer.
- ***Privacy***
- Because interactions with the physical world may often include humans, protecting the user privacy is of utmost importance for an IoT system.
- The IoT-A Privacy Model depends on the following functional components: Identity Management, Authentication, Authorization, and Trust & Reputation.

- ***Communication model***

- ***Trust***

- Generally, an entity is said to 'trust' a second entity when the first entity makes the assumption that the second entity will behave exactly as the first entity expects."

- ***Security***

- The Security Model for IoT consists of communication security that focuses mostly on the confidentiality and integrity protection of interacting entities and functional components such as Identity Management, Authentication, Authorization, and Trust & Reputation.

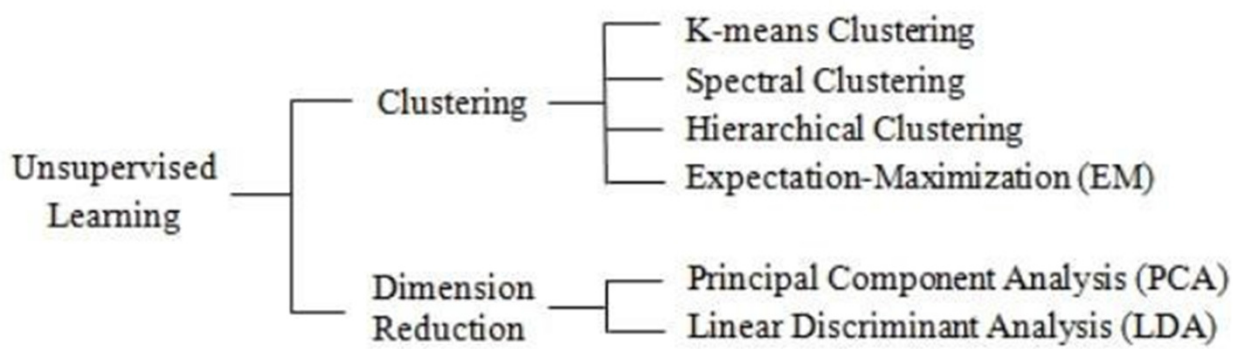
Taxonomy of machine learning algorithms

- The purpose of machine learning is to teach computers to execute tasks without human intervention.
- An increasing number of applications such as genomics, social networking, advertising, or risk analysis generate a very large amount of data that can be analyzed or mined to extract knowledge or insight into a process, customer, or organization.
- Ultimately, machine learning algorithms consist of identifying and validating models to optimize a performance criterion using historical, present, and future data.

Taxonomy of machine learning algorithms

- **Unsupervised learning**
- The goal of **unsupervised learning** is to discover patterns of regularities and irregularities in a set of observations.
- The process is known as density estimation in statistics is broken down into two categories: discovery of data clusters and discovery of latent factors.
- The methodology consists of processing input data to understand patterns similar to the natural learning process in infants or animals.
- Unsupervised learning does not require labeled data (or expected values), and therefore, it is easy to implement and execute because no expertise is needed to validate an output.
- However, it is possible to label the output of a clustering algorithm and use it for future classification.

Taxonomy of machine learning algorithms

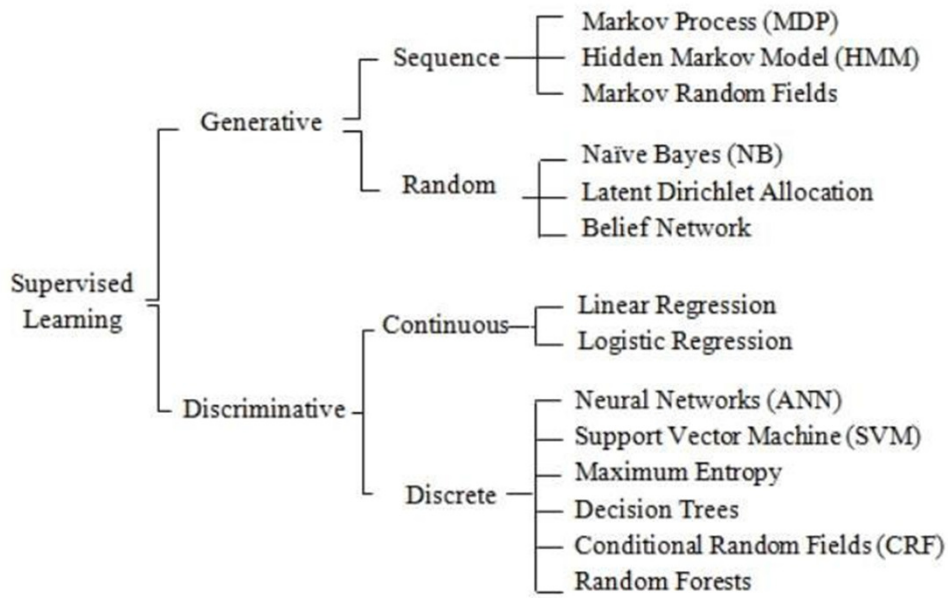


Taxonomy of unsupervised learning algorithms

Taxonomy of machine learning algorithms

- **Supervised learning**
- The best analogy for supervised learning is **function approximation** or **curve fitting**. In its simplest form, supervised learning attempts to find a relation or function **f: x → y** using a training set **{x, y}**.
- Supervised learning is far more accurate than any other learning strategy as long as the input (labeled data) is available and reliable.
- The downside is that a domain expert may be required to label (or tag) data as a training set.

Taxonomy of machine learning algorithms

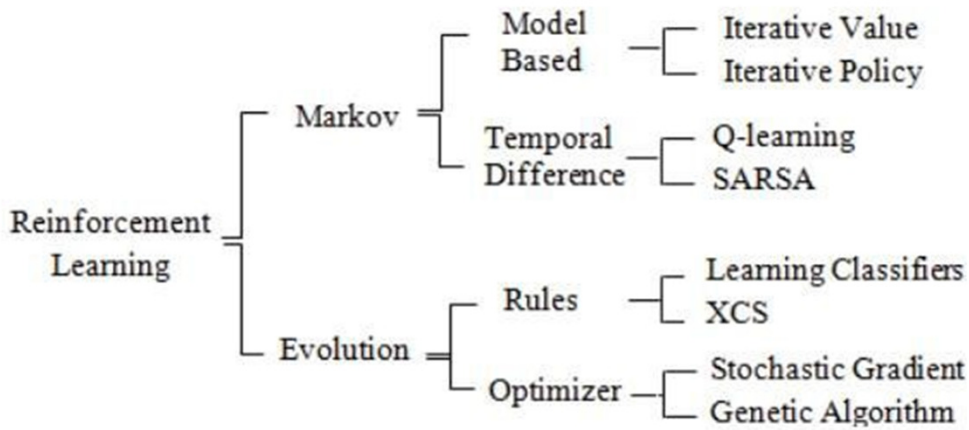


Taxonomy of supervised learning algorithms

Taxonomy of machine learning algorithms

- **Reinforcement learning**
- Reinforcement learning is not as well understood as supervised and unsupervised learning outside the realms of robotics or game strategy.
- However, since the 90s, genetic-algorithms-based classifiers have become increasingly popular to solve problems that require collaboration with a domain expert.
- For some types of applications, reinforcement learning algorithms output a set of recommended actions for the adaptive system to execute.
- In its simplest form, these algorithms estimate the best course of action.
- Most complex systems based on reinforcement learning establish and update policies that can be banned by an expert, if necessary.
- The foremost challenge developers of reinforcement learning systems face is that the recommended action or policy may depend on partially observable states.

Taxonomy of machine learning algorithms



Taxonomy of reinforcement learning algorithms

Taxonomy of machine learning algorithms

- **Semi-supervised learning**
- Semi-supervised learning is used to build models from a dataset with incomplete labels.
- Manifold learning and information geometry algorithms are commonly applied to large datasets that are partially labeled.

End
of
UNIT IV